

```
1 def lps_length(s):
2     for length in range(2, n + 1):
3         for i in range(n - length + 1):
4
5             if s[i] == s[j]:
6                 dp[i][j] = 2 + (dp[i + 1][j - 1] if length > 2
7                               else 0)
8             else:
9                 dp[i][j] = max(dp[i + 1][j], dp[i][j - 1])
10
11     return dp[0][n - 1]
12
13 print(lps_length("bbbab"))
14 print(lps_length("cbbd"))
15 print(lps_length(""))
```

```
4
2
0
```

=== Code Execution Successful ===

```
1 def palindrome_score(s):
12     for length in range(2, n + 1):
13         for i in range(n - length + 1):
15
16             if s[i] == s[j]:
17                 dp[i][j] = 2 + (dp[i + 1][j - 1] if length > 2
18                     else 0)
19             else:
20                 dp[i][j] = max(dp[i + 1][j], dp[i][j - 1])
21
22     return dp[0][n - 1]
23
24 print(palindrome_score("bbbab"))
25 print(palindrome_score("cbbd"))
26 print(palindrome_score("a"))
```

```
4
2
1
```

=== Code Execution Successful ===

```
1 def lps_length(s):
12     for length in range(2, n + 1):
13         for i in range(n - length + 1):
```

```
20
21     return dp[0][n - 1]
```

```
22
23
24 def symmetry_ratio(s):
25     if len(s) == 0:
26         return 0
27
28     return lps_length(s) / len(s)
```

```
29
30
31 print(round(symmetry_ratio("bbbab"), 2))
32 print(round(symmetry_ratio("cbbd"), 2))
```

0.8

0.5

=== Code Execution Successful ===

```
1 def lps_length(s):
12     for length in range(2, n + 1):
13         for i in range(n - length + 1):
14             if s[i] == s[i + length - 1]:
15                 dp[i][j] = dp[i][j] + 1
16             else:
17                 dp[i][j] = max(dp[i + 1][j], dp[i][j - 1])
18
19     return dp[0][n - 1]
20
21
22
23
24 def base_pair_estimate(s):
25     return lps_length(s) // 2
26
27
28 print(base_pair_estimate("bbbab"))
29 print(base_pair_estimate("cbbd"))
```

2
1
0
=== Code Execution Successful ===

```

1  def lps_length(s):
12     for length in range(2, n + 1):
13         for i in range(n - length + 1):
18             else:
19                 dp[i][j] = max(dp[i + 1][j], dp[i][j - 1])
20
21     return dp[0][n - 1]
22
23
24 def is_compression_candidate(s):
25     score = lps_length(s) - len(s) / 2
26     return score > 0
27
28
29 print(is_compression_candidate("bbbab"))
30 print(is_compression_candidate("cbbd"))

```

True

False

=== Code Execution Successful ===

```

1  def reconstruct_lps(s):
25     while i <= j:
30         if s[i] == s[j]:
34             elif dp[i + 1][j] >= dp[i][j - 1]:
35                 i += 1
36             else:
37                 j -= 1
38
39     left = result
40     right = result[::-1][::-1] if n % 2 == 1 else result[::-1]
41
42     return ''.join(left + right)
43
44
45 print(reconstruct_lps("bbbab"))
46 print(reconstruct_lps("cbbd"))

```

bbb

bb

=== Code Execution Successful ===

Power

Your bat
You sho

```
1 def lps_length_optimized(s):
9     for i in range(n - 2, -1, -1):
12         for j in range(i, n):
15             if j == i:
17                 elif s[i] == s[j]:
18                     dp[j] = prev + 2
19             else:
20                 dp[j] = max(dp[j], dp[j - 1])
21
22         prev = temp
23
24     return dp[n - 1]
25
26
27 print(lps_length_optimized("bbbab"))
28 print(lps_length_optimized("cbbd"))
```

```
5
3
```

```
=== Code Execution Successful ===
```

```
1 def lps_length(s):
23
24     return dp[n - 1]
25
26
27 def is_weak_password(password, threshold):
28     if len(password) == 0:
29         return False
30
31     ratio = lps_length(password) / len(password)
32
33     return ratio > threshold
34
35
36 print(is_weak_password("bbbab", 0.6))
37 print(is_weak_password("cbbd", 0.6))
```

True

True

=== Code Execution Successful ===


```
1 def lps_length(s):
21 |     return dp[0][n - 1]
22
23
24 def audit_logs(log, window_size):
25     result = []
26
27     for i in range(len(log) - window_size + 1):
28         window = log[i:i + window_size]
29         result.append(lps_length(window))
30
31     return result
32
33
34 print(audit_logs("bbbab", 5))
35 print(audit_logs("cbabd", 4))
```

[4]

[2]

=== Code Execution Successful ===

```

24 def lcs_length(a, b):
29
30     for i in range(1, m + 1):
31         for j in range(1, n + 1):
32             if a[i - 1] == b[j - 1]:
33                 dp[i][j] = dp[i - 1][j - 1] + 1
34             else:
35                 dp[i][j] = max(dp[i - 1][j], dp[i][j - 1])
36
37     return dp[m][n]
38
39
40 s = "bbbab"
41
42 print("LPS:", lps_length(s))
43 print("LCS:", lcs_length(s, s[::-1]))

```

LPS: 4

LCS: 4

=== Code Execution Successful ===